

JEEVANAA SURESH BABU

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PROFESSIONAL SUMMARY

Final-year EEE student at NIT Trichy with hands-on experience in industrial commissioning, electromechanical systems, and engineering simulation. Independently developed an LHC beam dynamics simulator using CERN's BLoND methodology, applying electrical engineering principles to accelerator systems. Adaptable and collaborative, with practical exposure to large-scale multidisciplinary engineering projects.

EDUCATION

National Institute of Technology Tiruchirappalli

B.Tech in Electrical and Electronics Engineering | **CGPA: 8.03**

Trichy, India

Aug 2023 – Present

Our Own English High School

CBSE Class 12: 92.8% | Class 10: 96.4%

Sharjah, UAE

Graduated 2023

PROJECTS

Longitudinal Phase Space Tracker for LHC Beam Dynamics

Nov 2025 – Jan 2026

- Developed a CERN-inspired longitudinal phase-space tracker implementing a **symplectic kick-drift algorithm** mirroring CERN's BLoND framework, using verified LHC injection parameters (**450 GeV, 6 MV RF voltage**).
- Simulated a **5,000-particle Gaussian proton bunch**; validated outputs against analytical predictions — reproducing **synchrotron frequency (56.49 Hz)**, **revolution period (88.92 μ s)**, and **RF bucket energy acceptance (268.86 MeV)**.
- Modelled phase-space evolution, RF separatrices, and beam stability; computed turn-by-turn RMS emittance via covariance-based diagnostics. *Tools: Python (NumPy, Matplotlib, SciPy), Jupyter Notebook.*

Laser-Based Line-of-Sight Communication System

Sept 2025 – Nov 2025

- Designed a **Free Space Optical (FSO)** audio transmission link with analog signal conditioning, multi-stage amplification, and laser intensity modulation for short-range (≤ 5 m) operation.
- Engineered optical receiver using a phototransistor and audio amplifier; analyzed signal integrity, noise sensitivity, and ambient light effects. *Tools: LTSpice, Proteus.*

INTERNSHIPS

Project Engineering Intern – AlManhad Electromechanical Engineering

June – July 2026

Multi-Aircraft Tail Dock and Nose Dock System

- Supported testing and commissioning of a Multi-Aircraft Tail Dock System — including **PLC/MCC panel wiring verification, loop testing, Megger (IR) testing**, and field commissioning within a 10-engineer team.
- Assisted in **sensor calibration and limit-setting** for the vertical lifting mechanism; structure achieved 91.6% of target operational height under test conditions.
- Independently developed a **3D CAD model** for a Multi-Aircraft Nose Dock concept, covering mechanism selection, structural layout, and cost estimation.

Astronomy & Astrophysics Intern – India Space Academy

June – July 2025

- Analysed archival spectroscopic and supernova datasets (Python, Astropy) to estimate galaxy cluster masses and the Hubble constant; performed image analysis using **SAOImage DS9**.

POSITIONS OF RESPONSIBILITY

Member, Nakshatra (Telescope Subteam) – Space & Astronomy Club, NIT Trichy

Jan 2025 – Jan 2026

- Contributed to assembly of a Schmidt-Cassegrain telescope and custom spectroscope; co-organised Star Dust (500+ participants) and a public outreach event for 200+ school students.

Additional Joint Secretary – Students' Council, NIT Trichy

July 2024 – April 2025

- Liaised between administration and 1000+ students; led initiatives including vending machines, hostel-wide Wi-Fi, and E-Vaahan services for a campus community of 5000+.

TECHNICAL SKILLS

Programming: Python, C, Verilog (HDL basics).

Engineering Software: MATLAB/Simulink, LTSpice, Proteus, Fusion 360 (CAD).

Core Skills: Power Electronics, SMPS, Control Systems, DSP, Analog & Digital Electronics, Microcontrollers.

Certifications: Electronic Systems Design – Hands-on Circuits & PCB with CAD (77%), MATLAB Onramp, Simulink Fundamentals.

Languages: Tamil (Native), English (Fluent), Hindi (Fluent), German (A1).